

Aymen Sadraoui, Ph.D.

Data Scientist & Data Engineer

Machine Learning, Clinical AI & Big Data Systems

Paris, France

aymensadraoui1919@gmail.com

+33 7 65 68 52 67

[Portfolio](#)

[LinkedIn](#)

Professional Summary

Data Scientist and Machine Learning Ph.D. graduate specialized in building robust **predictive models** and scalable **data engineering pipelines** for high-dimensional, multimodal datasets. I bring extensive hands-on expertise in **deep learning**, **representation learning**, and **few-shot learning**, maintaining a rigorous evaluation-first methodology focused on **model calibration**, **out-of-distribution generalization**, and reproducibility. Proven track record of translating complex mathematical research into first-author publications at top-tier venues (**CVPR**, **ISBI**, **ICIP**) and engineering production-grade **ETL pipelines** for clinical AI systems. I am seeking roles where I can architect end-to-end **data pipelines** and deploy scalable, impact-driven **AI/ML solutions**.

Technical Skills

Core Tech: Python, PyTorch, SQL, Linux, Git, AWS, CI/CD, Docker, MLflow, FastAPI, ETL/ELT Pipelines

Languages: English (Fluent), French (Fluent), Arabic (Native)

Professional Experience

Data Scientist & Machine Learning Engineer

Feb 2026 — Present

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Paris, France

- Architected and deployed end-to-end **automated data processing and ingestion pipelines** integrating directly with hospital PACS (DICOM) and cloud platforms for multi-centric clinical studies.
- Developed deep learning-based algorithms for **automated image quality control (QC)**, detecting motion artifacts, slice mismatches, and data corruption prior to clinical review.
- Engineered computer vision algorithms for **automated diagnosis, segmentation, and classification** of radiology scans (CT, X-ray, MRI) across orthopedics, oncology, and rheumatology modalities.
- Deployed scalable, containerized microservices (**Docker**, **FastAPI**, **Streamlite**, **PyTorch**) into production workflows, powering AI-driven pre-annotations and interactive 3D visualizations for clinical operators.
- Implemented robust **MLOps frameworks** for model calibration, and data drift detection to ensure high out-of-distribution generalization across multi-site hospital data.
- Partnered with cross-functional teams of radiologists, software engineers, and clinical managers to translate clinical research requirements into production-ready software features.

Ph.D. Researcher in AI & Computational Pathology

Jan 2022 — Jan 2026

CentraleSupélec (OPIS / CVN Lab) & Kremlin-Bicêtre Hospital

Gif-sur-Yvette, France

- Thesis:** *AI-Driven Methods in Histopathology for the Diagnosis and Prognosis of Hepatocellular Carcinoma.*
- Clinical-Technical Interface:** Served as the primary bridge between ML researchers and senior pathologists; translated qualitative histopathological criteria into quantifiable ML objectives and validated automated features against clinical consensus.
- Advanced Deep Learning & Optimization:** Designed and unrolled proximal optimization algorithms (Unrolled Projected Gradient, UNEM) into neural networks for robust stain separation, color normalization, and transductive few-shot slide classification.
- Terabyte-Scale Data Engineering:** Built scalable ETL preprocessing pipelines (PyTorch, OpenCV, OpenSlide, Ray) to ingest, tile, and curate over 4+ TB of gigapixel whole slide images, handling high-throughput data processing and multi-source artifact removal.
- Multi-Modal Predictive Modeling:** Developed deep learning models integrating histopathologic architecture quantification with clinical/prognostic clinical metadata to predict early liver cancer recurrence.
- MLOps & Trustworthy AI:** Created rigorous benchmarking frameworks focused on out-of-distribution generalization, model calibration, uncertainty estimation, and interpretability for high-stakes medical deployment.

Achievements

- 1st Place Winner:** Lung Transplant Rejection Challenge — Organized by Hôpital Foch & Health Data Hub (2024)
- National Rank 63 / 1,773 (Top 3%):** National Engineering Entrance Exam, Tunisia (2018)

Education

Ph.D. in Machine Learning & Artificial Intelligence

2022 — 2026

CentraleSupélec, Université Paris-Saclay

Gif-sur-Yvette, France

Engineering Degree in Signals and Systems Processing

2018 — 2021

École Polytechnique de Tunisie

La Marsa, Tunisia

Publications

- **Sadraoui, A., et al.** (2026). A Proximal Approach for Stain Separation and Normalization of Whole-Slide Histopathological Images. *IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*.
- **Sadraoui, A., et al.** (2025). UNEM: UNrolled Generalized EM for Transductive Few-Shot Learning. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- **Sadraoui, A., et al.** (2024). Deep Learning Classification and Quantification of Pejorative and Nonpejorative Architectures in Resected Hepatocellular Carcinoma from Digital Histopathologic Images. *The American Journal of Pathology*
- **Sadraoui, A., et al.** (2024). Unrolled Projected Gradient Algorithm for Stain Separation in Digital Histopathological Images. *IEEE International Conference on Image Processing (ICIP)*
- **Sadraoui, A., et al.** (2024). A Transductive Few-Shot Learning Approach for Classification of Digital Histopathological Slides from Liver Cancer. *IEEE International Symposium on Biomedical Imaging (ISBI)*